1	2	3	4	5	6	Total
/20	/20	/20	/20	/20	/20	/100

Name: ______________________________ June 23, 2019. Sunday

Department/School:____________________ SID:_________________________________

University Physics

Final Examination

Kuang Yaming Honors School, Nanjing University

$$\Gamma(\nu)=\int_0^{+\infty} e^{-x}x^{\nu-1}dx,\quad \Gamma(\nu+1)=\nu\Gamma(\nu),\quad \Gamma(1)=1,\quad \Gamma(1/2)=\sqrt{\pi}.$$

Select five out of following six problems.

1.(20pts.) There happen two events that are at the same place and a time interval $\Delta t' = 300\text{s}$ apart in the S' reference frame. In the S reference frame, the two events are $\Delta t = 500\text{s}$ apart. The S' frame moves with speed u relative to S frame along the common x and x' direction.

(a) What is the speed u ?

(b) What is the space interval d between two events in the S frame?

Solution: (a) From $\Delta t = \gamma\left(\Delta t' + \dfrac{u}{c^2}\Delta x'\right)$, $\Delta x' = 0$, $\Delta t' = 300\text{s}$, $\Delta t = 500\text{s}$, we have

$$\frac{1}{\gamma} = \sqrt{1-\left(\frac{u}{c}\right)^2} = \frac{\Delta t'}{\Delta t} = \frac{3}{5},\quad \frac{u}{c} = \frac{4}{5},\quad u = \frac{4}{5}c\,.$$

(b) $$d = \Delta x = \gamma(\Delta x' + u\Delta t') = \frac{5}{3}u\Delta t' = \frac{4}{3}c\Delta t' = 1.2\times 10^{11}\ \text{m}\,.$$

2. (20pts.) A block of ice of refraction index n floats on the water of a river and travels with speed v. A beam of light propagates through the transparent ice block in the same direction of the velocity of the moving ice.

(a) What is the speed of the light v_1 in the ice to the observer that travels with the ice block?

(b) What is the speed of the light v_2 in the ice to the observer standing on the shore of the river?

Solution: (a) $v_1 = c/n$.

(b) $$v_2 = \frac{v_1 + v}{1+\dfrac{v_1 v}{c^2}} = \frac{\dfrac{c}{n}+v}{1+\dfrac{v}{cn}}\,.$$

3.(20pts.) A block of ice of mass 2 kg and at temperature of 0°C is placed into a bath of water of 10 kg at temperature 10 ℃. The ice and the bath of water as a whole are adiabatically enveloped. After some time, the system is in equilibrium again. The latent heat of melting of ice (at 0°C) is $3.35\times10^5\text{ J/kg}$ and the specific heat capacity of water at constant pressure is $4.187\times10^3\text{ J/kg}\cdot\text{K}$.

(a) Find ΔS_1, the change in entropy of the ice .

(b) Find ΔS_2, the change in entropy of the bath of water.

(c) Find ΔS_t, the change in entropy of the total system.

Solution: (a) $m = 2\text{ kg}$, $L = 3.35\times10^5\text{ J/kg}$, $c_p = 4.187\times10^3\text{ J/kg}\cdot\text{K}$, $T_1 = 273.15\text{ K}$, $T_2 = 283.15\text{ K}$.

$mL = 6.7\times10^5\text{ J}$, $m_2c_p(T_2 - T_1) = 4.187\times10^5\text{ J}$

Since $m_2c_p(T_2 - T_1) < mL$, only part of ice melts and the final equilibrium is a mixture of ice and water at $T = 0^\circ\text{C}$.
Thus the change in entropy of the ice is

$$\Delta S_1 = \frac{m'L}{T_1} = \frac{m_2c_p(T_2 - T_1)}{T_1} = 1.53\times10^3\text{ J/K}.$$

(b)

$$\Delta S_2 = \int_{T_2}^{T_1}\frac{m_2c_p\,dT}{T} = -m_2c_p\ln\left(\frac{T_2}{T_1}\right) = -1.51\times10^3\text{ J/K}.$$

(c) $$\Delta S_t = \Delta S_1 + \Delta S_2 = 0.2\times10^2\text{ J/K}.$$

4. (20pts.) A system of gas of fixed number of molecules is with two thermodynamic degrees of freedom (or two independent thermodynamic coordinates) only. The volume V and the internal energy U can be chosen as the two thermodynamic degrees of freedom (or thermodynamic coordinates).

(a) Write down the differential form of the temperature T with respects of volume V and internal energy U.

(b) Show from (a) that $dU = \left[\left(\frac{\partial T}{\partial U}\right)_V\right]^{-1} dT - \left[\left(\frac{\partial T}{\partial U}\right)_V\right]^{-1}\left(\frac{\partial T}{\partial V}\right)_U dV$.

(c) Show that, if it were found in the Joule's experiment of free expansion of the gas that $\left(\frac{\partial T}{\partial V}\right)_U = 0$, the internal energy U of the gas could be the function of temperature T only.

(d) We all know that the internal energy of the ideal gas is the function of temperature T only. Then, if a mole of ideal gas freely expands from v_1 to v_2 as in the Joule's experiment, what is the change in entropy Δs for the gas?

Solution: (a)

$$dT = \left(\frac{\partial T}{\partial V}\right)_U dV + \left(\frac{\partial T}{\partial U}\right)_V dU .$$

(b)

$$\left(\frac{\partial T}{\partial U}\right)_V dU = dT - \left(\frac{\partial T}{\partial V}\right)_U dV \qquad dU = \left[\left(\frac{\partial T}{\partial U}\right)_V\right]^{-1} dT - \left[\left(\frac{\partial T}{\partial U}\right)_V\right]^{-1}\left(\frac{\partial T}{\partial V}\right)_U dV .$$

(c) Substituting $\left(\frac{\partial T}{\partial V}\right)_U = 0$ into above equation, we obtain

$$dU = \left[\left(\frac{\partial T}{\partial U}\right)_V\right]^{-1} dT .$$

Thus, the internal energy is the function of temperature T only.

(b) In the free expansion, $dW = 0$, $dQ = 0$, then $dU = 0$. Therefore, from the fundamental equation of thermodynamics $dU = TdS - pdV$, we have

$$dS = \frac{p}{T} dV .$$

For a mole of ideal gas, we have

$$ds = \frac{p}{T} dv = \frac{\frac{RT}{v}}{T} dv = \frac{R}{v} dv .$$

Finally

$$\Delta s = \int_{v_1}^{v_2} \frac{R}{v} dv = R \ln \frac{v_2}{v_1} .$$

5. (20 pts.) Let U, F, S, T, p, V and C_V denote the internal energy, Helmholtz free energy, entropy, temperature, pressure, volume, and heat capacity at constant volume of a gas of fixed particles, respectively.

(a) Show that $TdS = C_V dT - T\left(\partial^2 F/\partial T\partial V\right)dV$.

(b) Show that $\left(\partial V/\partial T\right)_S = -\left(C_V/T\right)\left(\partial T/\partial p\right)_V$.

(c) Show that $\left(\partial U/\partial p\right)_V = -T\left(\partial V/\partial T\right)_S$.

Solution: (a) From $dU = TdS - pdV$, we have

$$dU = T\left[\left(\frac{\partial S}{\partial T}\right)_V dT + \left(\frac{\partial S}{\partial V}\right)_T dV\right] - pdV = T\left(\frac{\partial S}{\partial T}\right)_V dT + \left[T\left(\frac{\partial S}{\partial V}\right)_T - p\right]dV .$$

Thus, we have $C_V = T\left(\frac{\partial S}{\partial T}\right)_V$ or $\left(\frac{\partial S}{\partial T}\right)_V = \frac{C_V}{T}$.

Using the Legendre transformation, we have

$$dF = d(U - TS) = -SdT - pdV .$$

Therefore, we have $S(T,V) = -\left(\frac{\partial F}{\partial T}\right)_V$ and $\left(\frac{\partial S}{\partial V}\right)_T = -\left[\frac{\partial}{\partial V}\left(\frac{\partial F}{\partial T}\right)_V\right]_T = -\frac{\partial^2 F}{\partial T\partial V}$.

Finally, $TdS = T\left(\frac{\partial S}{\partial T}\right)_V dT + T\left(\frac{\partial S}{\partial V}\right)_T dV = C_V dT - T\frac{\partial^2 F}{\partial T\partial V}dV$.

(b) From $dF = -SdT - pdV$, we have the Maxwell relation $\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial p}{\partial T}\right)_V$.

Thus $TdS = T\left[\left(\frac{\partial S}{\partial T}\right)_V dT + \left(\frac{\partial S}{\partial V}\right)_T dV\right] = C_V dT + T\left(\frac{\partial p}{\partial T}\right)_V dV$, or

$$dV = \frac{1}{\left(\frac{\partial p}{\partial T}\right)_V} dS - \frac{C_V}{T\left(\frac{\partial p}{\partial T}\right)_V} dT .$$

Therefore $\left(\frac{\partial V}{\partial T}\right)_S = -\frac{C_V}{T\left(\frac{\partial p}{\partial T}\right)_V} = -\frac{C_V}{T}\left(\frac{\partial T}{\partial p}\right)_V$.

(c) From $dU = C_V dT + \left[T\left(\frac{\partial p}{\partial T}\right)_V - p\right]dV$ and $dT = \left(\frac{\partial T}{\partial p}\right)_V dp + \left(\frac{\partial T}{\partial V}\right)_p dV$, we have

$$dU = C_V\left(\frac{\partial T}{\partial p}\right)_V dp + \left[C_V\left(\frac{\partial T}{\partial V}\right)_p + T\left(\frac{\partial p}{\partial T}\right)_V - p\right]dV .$$

Thus $\left(\frac{\partial U}{\partial p}\right)_V = C_V\left(\frac{\partial T}{\partial p}\right)_V$.

From (b) $\left(\frac{\partial V}{\partial T}\right)_S = -\frac{C_V}{T}\left(\frac{\partial T}{\partial p}\right)_V$, we obtain

$$\left(\frac{\partial U}{\partial p}\right)_V = -T\left(\frac{\partial V}{\partial T}\right)_S .$$

6. (20pts.) From the kinetic theory of gases, the pressure of a gas can be determined by the average of the quantity $nmv^2/3$, where n is the number density of molecules, m is the mass of the molecule and v is the speed of the molecule.

(a) Find the pressure of the gas in terms of temperature T by applying the Maxwell distribution.

(b) Find the fluctuation (or uncertainty) of the pressure Δp. ($\Delta p = \sqrt{\left\langle \left(nmv^2/3\right)^2 \right\rangle - \left\langle nmv^2/3 \right\rangle^2}$)

Solution: (a)

$$p = \left\langle \frac{nmv^2}{3} \right\rangle = \frac{\int \frac{nmv^2}{3} dn(v)}{\int dn(v)} = \frac{nm}{3} \frac{\int_0^{+\infty} v^2 v^2 e^{-\frac{mv^2}{2k_BT}} dv}{\int_0^{+\infty} v^2 e^{-\frac{mv^2}{2k_BT}} dv} \underset{\frac{mv^2}{2k_BT}=x^2}{=} \frac{nm}{3} \frac{2k_BT}{m} \frac{\int_0^{+\infty} x^4 e^{-x^2} dx}{\int_0^{+\infty} x^2 e^{-x^2} dx}$$

$$\underset{x^2=t}{=} \frac{2}{3} nk_BT \frac{\int_0^{+\infty} t^{3/2} e^{-t^2} dt}{\int_0^{+\infty} t^{1/2} e^{-t} dt} = \frac{2}{3} nk_BT \frac{\Gamma\left(\frac{5}{2}\right)}{\Gamma\left(\frac{3}{2}\right)} = \frac{2}{3} nk_BT \frac{\frac{3}{2}\Gamma\left(\frac{3}{2}\right)}{\Gamma\left(\frac{3}{2}\right)} = nk_BT$$

$$= \frac{Nk_BT}{V} = \frac{n_m N_A k_B T}{V} = \frac{n_m RT}{V}.$$

(b)

$$\left\langle \left(\frac{nmv^2}{3}\right)^2 \right\rangle = \frac{\int \left(\frac{nmv^2}{3}\right)^2 dn(v)}{\int dn(v)} = \frac{n^2m^2}{9} \frac{\int_0^{+\infty} v^4 v^2 e^{-\frac{mv^2}{2k_BT}} dv}{\int_0^{+\infty} v^2 e^{-\frac{mv^2}{2k_BT}} dv} \underset{\frac{mv^2}{2k_BT}=x^2}{=} \frac{n^2m^2}{9} \left(\frac{2k_BT}{m}\right)^2 \frac{\int_0^{+\infty} x^6 e^{-x^2} dx}{\int_0^{+\infty} x^2 e^{-x^2} dx}$$

$$\underset{x^2=t}{=} \frac{4}{9}\left(nk_BT\right)^2 \frac{\int_0^{+\infty} t^{5/2} e^{-t^2} dt}{\int_0^{+\infty} t^{1/2} e^{-t} dt} = \frac{4}{9}\left(nk_BT\right)^2 \frac{\Gamma\left(\frac{7}{2}\right)}{\Gamma\left(\frac{3}{2}\right)} = \frac{4}{9}\left(nk_BT\right)^2 \frac{\frac{5}{2}\frac{3}{2}\Gamma\left(\frac{3}{2}\right)}{\Gamma\left(\frac{3}{2}\right)} = \frac{5}{3}\left(nk_BT\right)^2.$$

$$\Delta p = \sqrt{\left\langle \left(\frac{nmv^2}{3}\right)^2 \right\rangle - \left\langle \frac{nmv^2}{3} \right\rangle^2} = \sqrt{\frac{2}{3}} nk_BT .$$